

$$\hat{y} = X_m + b, \text{ where } \hat{y}_i = m_1 x_{i1} + m_2 x_{i2} + b_i$$

Given :

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad m^0 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad b^0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Cost function (MSE), $n = 2$; data points

$$J(m, b) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Learning rate $\alpha = 0.01$

$$\text{update rules: } m \leftarrow m - \alpha \frac{\partial J}{\partial m} \quad \text{and} \quad b \leftarrow b - \alpha \frac{\partial J}{\partial b}$$

Deriving the gradient with the chain rule.

let $e = \bar{y} - y$ (error vector), so $J = \frac{1}{n} \sum_i e_i^2$

Differentiate through the chain: $J \rightarrow e \rightarrow \bar{y} \rightarrow (m, b)$

Step 1 (outer): $\frac{\partial J}{\partial e_i} = \frac{\partial}{\partial e_i} \left(\frac{1}{n} \sum_k e_k^2 \right) = \frac{2}{n} e_i$

Step 2 (inner): since $\bar{y}_i = m_1 x_{i1} + m_2 x_{i2} + b_i$

$$\frac{\partial e_i}{\partial \bar{y}_i} = 1$$

Step 3 (outer): since $\bar{y}_i = m_1 x_{i1} + m_2 x_{i2} + b_i$

$$\frac{\partial \bar{y}_i}{\partial m_j} = x_{ij} \quad \frac{\partial \bar{y}_i}{\partial b_i} = 1$$

Combine - chain rule.

For m_j (sum over i because every prediction depends on m_j)

$$\frac{\partial J}{\partial m_j} = \sum_{i=1}^n \frac{\partial J}{\partial e_i} \cdot \frac{\partial e_i}{\partial \bar{y}_i} \cdot \frac{\partial \bar{y}_i}{\partial m_j}$$

$$= \sum_{i=1}^n \frac{2}{n} e_i \cdot 1 \cdot x_{ij} = \frac{2}{n} \sum_{i=1}^n x_{ij} e_i$$

$$= \boxed{\frac{\partial J}{\partial m} = \frac{2}{n} X^T e}$$

For b_i (only its own prediction depends on b_i):

$$\frac{\partial J}{\partial b_i} = \frac{2}{n} e_i \cdot 1 \cdot 1 = \boxed{\frac{\partial J}{\partial b} = \frac{2}{n} e}$$

Since $n=2$: $\frac{2}{n} = \frac{2}{2} = 1$;

$$\therefore \frac{\partial J}{\partial m} = X^T e \text{ and } \frac{\partial J}{\partial b} = e$$

Iteration 1 : (Merai)

$$\text{start } m^{(0)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}; \quad b^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

a) Predictions:

$$\hat{y}_1 = (1)(-1) + (3)(2) + 1 = -1 + 6 + 1 = 6$$

$$\hat{y}_2 = (4)(-1) + (10)(2) + 1 = -4 + 20 + 1 = 17$$

$$\text{b) Errors \& cost: } e = \begin{bmatrix} 6 - 5 \\ 17 - 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \frac{(1)^2 + 11^2}{2} = \frac{1 + 121}{2} = \frac{122}{2}$$

= 61.000

$$\text{c) Gradient: } \frac{\partial J}{\partial m} = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} (1)(1) + (4)(11) \\ (3)(1) + (10)(11) \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 44 \\ 3 + 110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = e = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

d) Updates ($\alpha = 0.01$)

$$m^{(1)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1 - 0.45 \\ 2 - 1.13 \end{bmatrix}$$

$$b^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 - 0.01 \\ 1 - 0.11 \end{bmatrix} = \begin{bmatrix} 0.9900 \\ 0.8900 \end{bmatrix}$$

Iteration Bias (1 van)

$$\text{start: } m^{(1)} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \quad b^{(1)} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\text{a) Predictions: } \hat{y}_1 = (1)(-1.45) + (3)(0.87) + 0.99 = -1.45 + 2.61 + 0.99 = 2.1500$$

$$\hat{y}_2 = (4)(-1.45) + (10)(0.87) + 0.89 = -5.80 + 8.70 + 0.89 = 3.7900$$

$$\text{b) Errors \& cost: } e = \begin{bmatrix} 2.1500 - 5 \\ 3.7900 - 6 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$J = \frac{(-2.85)^2 + (-2.21)^2}{2} = \frac{8.1225 + 4.8841}{2} = 6.5033$$

$$\text{c) Gradient: } \frac{\partial J}{\partial m} = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} (1)(-2.85) + (4)(-2.21) \\ (3)(-2.85) + (10)(-2.21) \end{bmatrix}$$

$$= \begin{bmatrix} -11.6900 \\ -30.6500 \end{bmatrix} = \frac{\partial J}{\partial b} = e = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$\text{d) Updates: } m^{(2)} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.001 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix} = \begin{bmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{bmatrix} = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}$$

$$b^{(2)} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.001 \begin{bmatrix} 2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} 0.99 + 0.0285 \\ 0.89 + 0.0221 \end{bmatrix} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

Iteration 3 (Dorian)

$$\text{Start } m^2 = \begin{bmatrix} 1.3331 \\ 1.1765 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

$$\text{a) Predictions: } \hat{y}_1 = (1)(-1.3331) + (3)(1.1765) + 1.0185 = -1.3331 + 3.5295 + 1.0185 \\ = 3.2149$$

$$\hat{y}_2 = (4)(-1.3331) + (10)(1.1765) + 0.9121 = -5.3324 + 11.765 + 0.9121 \\ = 7.3447$$

$$\text{b) Errors \& cost: } e = \begin{bmatrix} 3.2149 - 5 \\ 7.3447 - 6 \end{bmatrix} = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

$$J = \frac{(-1.7851)^2 + (1.3447)^2}{2} = \frac{3.1866 + 1.8082}{2} = \underline{\underline{2.4974}}$$

$$\text{Gradients: } \frac{\partial J}{\partial m} = x^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} = \begin{bmatrix} (1)(-1.7851) + (4)(1.3447) \\ (3)(-1.7851) + (10)(1.3447) \end{bmatrix}$$

$$\begin{bmatrix} -1.7851 + 5.3788 \\ -5.3553 + 13.4470 \end{bmatrix} = \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix}; \quad \frac{\partial J}{\partial b} = e = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

Updates:

$$m^{(3)} = \begin{bmatrix} 1.3331 \\ 1.1765 \end{bmatrix} - 0.01 \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix} = \begin{bmatrix} 1.3331 - 0.035937 \\ 1.1765 - 0.080917 \end{bmatrix} = \begin{bmatrix} 1.297163 \\ 1.095583 \end{bmatrix}$$

$$b^{(3)} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - 0.01 \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} = \begin{bmatrix} 1.0185 + 0.017851 \\ 0.9121 - 0.013447 \end{bmatrix} = \begin{bmatrix} 1.036351 \\ 0.898653 \end{bmatrix}$$

Iteration 4 (Belyse)

$$\text{Start } m^{(3)} = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} \quad b^3 = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

a) Predictions:

$$\hat{y}_1 = (1)(-1.3690) + (3)(1.0956) + 1.0364 = -1.3690 + 3.2968 + 1.0364 = 2.9542$$

$$\hat{y}_2 = (4)(-1.3690) + (10)(1.0956) + 0.8987 = -5.4760 + 10.9560 + 0.8987 = 6.3787$$

$$b) \text{ Errors \& cost; } e = \begin{bmatrix} 2.9542 - 5 \\ 6.3787 - 6 \end{bmatrix} = \begin{bmatrix} -2.0458 \\ 0.3787 \end{bmatrix}$$

$$J = \frac{(-2.0458)^2 + (0.3787)^2}{2} = \frac{4.1853 + 0.1434}{2} = 2.1644$$

$$c) \text{ Gradients: } \frac{\partial J}{\partial m} = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.0458 \\ 0.3787 \end{bmatrix}$$

$$\begin{bmatrix} (1)(-2.0458) + (4)(0.3787) \\ (3)(-2.0458) + (10)(0.3787) \end{bmatrix} = \begin{bmatrix} -2.0458 + 1.5148 \\ -6.1374 + 3.7870 \end{bmatrix} = \begin{bmatrix} -0.5310 \\ -2.3504 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = e = \begin{bmatrix} -2.0458 \\ 0.3787 \end{bmatrix}$$

Updates:

$$m^4 = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} - 0.01 \begin{bmatrix} -0.5310 \\ -2.3504 \end{bmatrix} = \begin{bmatrix} -1.3690 + 0.005310 \\ 1.0956 + 0.023504 \end{bmatrix} = \begin{bmatrix} -1.3637 \\ 1.1191 \end{bmatrix}$$

$$b^4 = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix} - 0.01 \begin{bmatrix} -2.0458 \\ 0.3787 \end{bmatrix} = \begin{bmatrix} 1.0364 + 0.020458 \\ 0.8987 + 0.003787 \end{bmatrix} = \begin{bmatrix} 1.0569 \\ 0.8949 \end{bmatrix}$$

Summary Table.

Iter	m_1	m_2	b_1	b_2	J(MSE)
0	-1.0000	2.0000	1.0000	1.0000	61.0000
1	-1.4500	0.8700	0.9900	0.9900	6.5033
2	-1.3331	1.1765	1.0185	0.9121	2.4974
3	-1.3690	1.0956	1.0364	0.8987	2.1044
4	-1.3637	1.1191	1.0569	0.8949	2.0996

Yes, m and b are moving in a direction that reduces the error. The MSE decreases at every iteration: $61.00 \rightarrow 6.50 \rightarrow 2.50 \rightarrow 2.16 \rightarrow 2.10$;

The largest drop happens in iteration 1, because at the start our predictions were very far from the real values, so the model had a lot to correct. After that, the changes to m and b became smaller & smaller, because the closer we get to the best values, the smaller the gradients become. This shows the gradient descent is working as expected.